torch.sparse.sum#

https://pytorch.org/docs/stable/generated/torch.sparse.sum.html

Description:

Return the sum of each row of the given sparse tensor. Returns the sum of each row of the sparse tensor input in the given dimensions dim. If dim is a list of dimensions, reduce over all of them. When sum over all sparse_dim, this method returns a dense tensor instead of a sparse tensor. All summed dim are squeezed (see torch.squeeze()), resulting an output tensor having dim fewer dimensions than input. During backward, only gradients at nnz locations of input will propagate back. Note that the gradients of input is coalesced. input (Tensor) – the input sparse tensor

Function Signature

```
torch.sparse.sum(input, dim=None, dtype=None)[source]#
Parameters
Return type
```

Returns:

Tensor

Code Examples

```
>>> nnz = 3
>>> dims = [5, 5, 2, 3]
>>> I = torch.cat([torch.randint(0, dims[0], size=(nnz,)),
                   torch.randint(0, dims[1], size=(nnz,))],
0).reshape(2, nnz)
>>> V = torch.randn(nnz, dims[2], dims[3])
>>> size = torch.Size(dims)
>>> S = torch.sparse_coo_tensor(I, V, size)
>>> S
tensor(indices=tensor([[2, 0, 3],
                       [2, 4, 1]]),
       values=tensor([[[-0.6438, -1.6467,  1.4004],
                       [ 0.3411,  0.0918, -0.2312]],

                      [[ 0.5348,  0.0634, -2.0494],
                       [-0.7125, -1.0646,  2.1844]],

                      [[ 0.1276,  0.1874, -0.6334],
                       [-1.9682, -0.5340,  0.7483]]]),
       size=(5, 5, 2, 3), nnz=3, layout=torch.sparse_coo)

# when sum over only part of sparse_dims, return a sparse tensor
>>> torch.sparse.sum(S, [1, 3])
tensor(indices=tensor([[0, 2, 3]]),
       values=tensor([[-1.4512,  0.4073],
                      [-0.8901,  0.2017],
                      [-0.3183, -1.7539]]),
       size=(5, 2), nnz=3, layout=torch.sparse_coo)

# when sum over all sparse dim, return a dense tensor
# with summed dims squeezed
>>> torch.sparse.sum(S, [0, 1, 3])
tensor([-2.6596, -1.1450])
```

Detailed Documentation

Documentation

Return the sum of each row of the given sparse tensor.

Returns the sum of each row of the sparse tensor input in the given dimensions dim. If dim is a list of dimensions, reduce over all of them. When sum over all sparse_dim, this method returns a dense tensor instead of a sparse tensor.

All summed dim are squeezed (see torch.squeeze()), resulting an output tensor having dim fewer dimensions than input.

During backward, only gradients at nnz locations of input will propagate back. Note that the gradients of input is coalesced.

input (Tensor) – the input sparse tensor

dim (int or tuple of ints) – a dimension or a list of dimensions to reduce. Default: reduce over all dims.

dtype (torch.dtype, optional) – the desired data type of returned Tensor. Default: dtype of input.